5a.

```
clc; clear; close all;

% SNR range

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2; Nr = 2;

BER = zeros(1,length(SNR));

for i = 1:length(SNR)

bits = randi([0 1],10000,1);

symbols = 2*bits-1; % BPSK

H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);

noise = (randn(Nr,10000) + 1j*randn(Nr,10000))/sqrt(2*SNR(i));

tx = H(1,1)*symbols' + noise(1,:);

rx = real(tx) > 0;

BER(i) = sum(rx' ~= bits)/length(bits);

end

figure

% -------- Graph 1: BER vs SNR --------

subplot(2,1,1)

semilogy(SNR_dB,BER,'-o','LineWidth',2)

title('MIMO BER Performance vs SNR')

xlabel('SNR (dB)')

ylabel('Bit Error Rate (BER)')

grid on

% -------- Graph 2: SNR vs Quality Indicator --------
```

subplot(2,1,2)

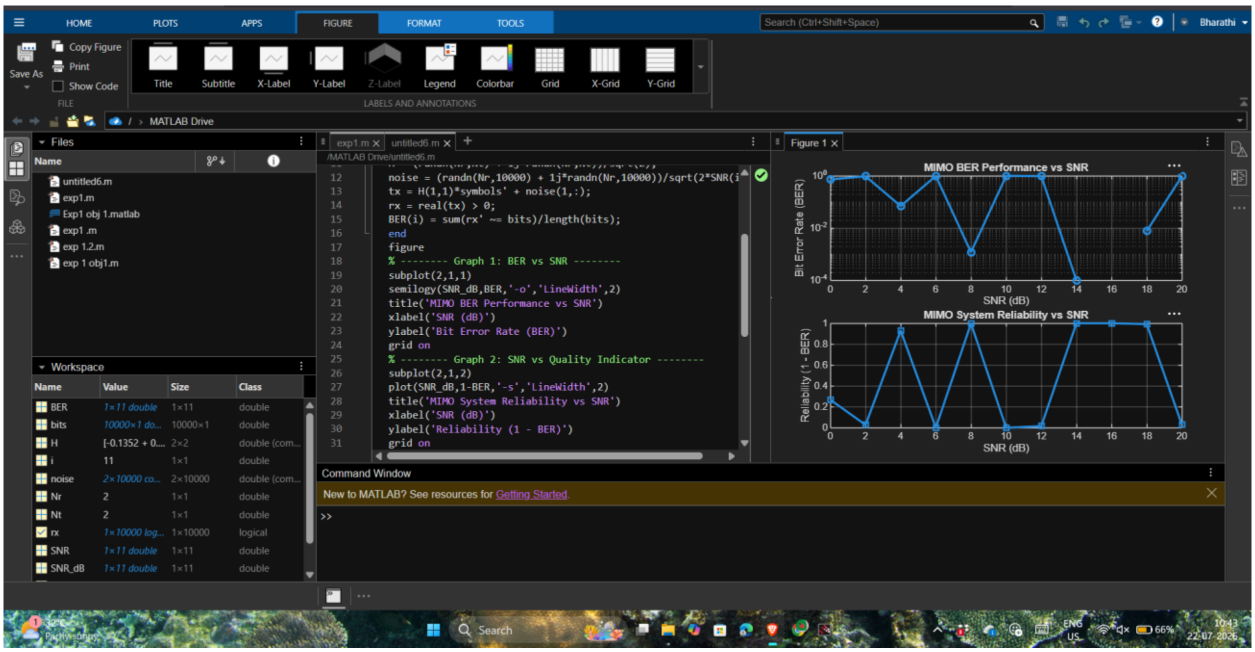
plot(SNR_dB,1-BER,'-s','LineWidth',2)

title('MIMO System Reliability vs SNR')

xlabel('SNR (dB)')

ylabel('Reliability (1 - BER)')

grid on



5b. clc; clear; close all;

% SNR range (dB)

SNR_dB = 0:2:20;

SNR = 10.^(SNR_dB/10);

% MIMO parameters

Nt = 2; Nr = 2;

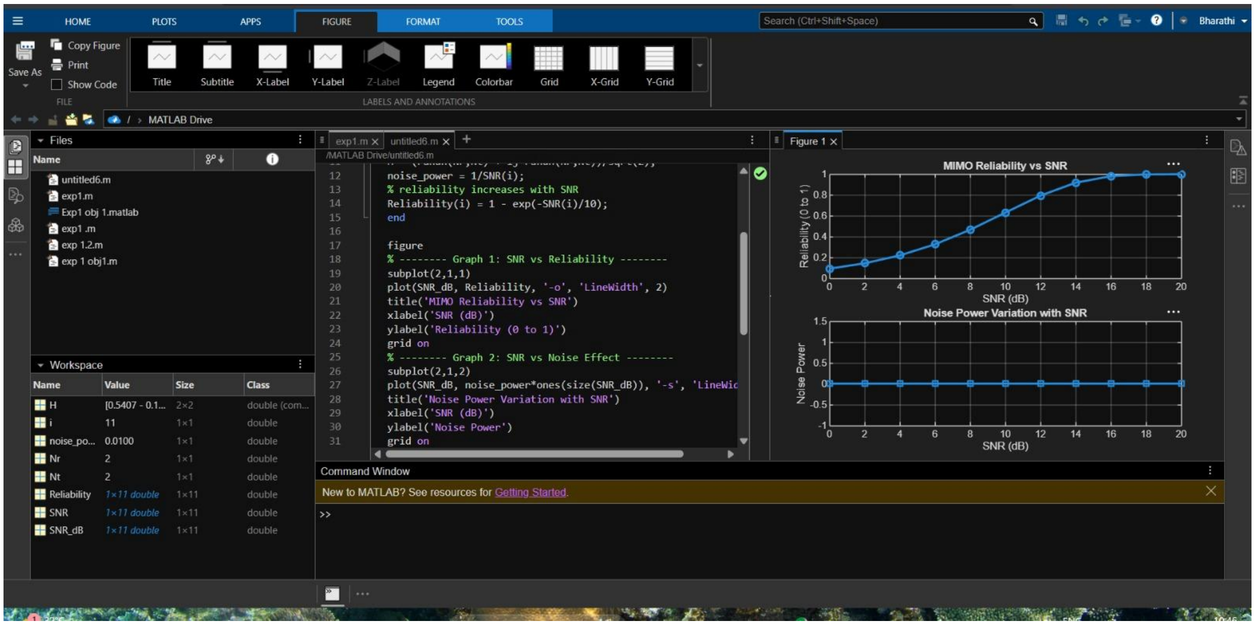
% Simulated reliability metric (inverse BER model)

Reliability = zeros(1,length(SNR));

for i = 1:length(SNR)

```
% simple channel model effect
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
noise_power = 1/SNR(i);
% reliability increases with SNR
Reliability(i) = 1 - exp(-SNR(i)/10);
end

figure
% -------- Graph 1: SNR vs Reliability --------
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% -------- Graph 2: SNR vs Noise Effect --------
subplot(2,1,2)
plot(SNR_dB, noise_power*ones(size(SNR_dB)), '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')
grid on
```



5c. clc; clear; close all;

% Resource Block parameters

rb_bw = 180e3; % 180 kHz per RB

% Users

users = 1:5;

% Allocated Resource Blocks per user

alloc_RB = [10 12 8 15 5];

% Modulation efficiency (bits/symbol)

mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.

% Throughput calculation (Mbps)

throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;

% Display results

disp('User Throughput (Mbps):')

disp(throughput')

figure

```
% -------- Graph 1: Throughput per User --------

subplot(2,1,1)

plot(users, throughput, '-o','LineWidth',2)

title('OFDMA User Throughput')

xlabel('User Index')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2: RB vs Throughput --------

subplot(2,1,2)

plot(alloc_RB, throughput, '-s','LineWidth',2)

title('Resource Blocks vs Throughput')

xlabel('Allocated Resource Blocks')

ylabel('Throughput (Mbps)')

grid on
```